PART 2

Airway and Breathing Emergencies

IN THIS PART

1. Identify the structures of the upper airway and what narrows or blocks them.
2. Recognize partial and complete airway obstruction and state the response.
3. Describe basic airway positioning, suction and bag-valve-mask ventilation, and their safety limits.
4. Recognize respiratory distress and respiratory failure and state first priorities for asthma, COPD and pneumonia.
5. Recognize anaphylaxis and state the first priorities.

A patient can survive a long time without food and a good while without a heartbeat if someone is pumping the chest. Without air reaching the lungs, the margin is measured in minutes. That is why airway and breathing come first in the primary survey of Entry 1.3, and why this Part comes next.

The five entries follow the path that air takes and the ways that path fails. Entry 2.1 describes the passage itself and how to judge whether it is open. Entry 2.2 covers what to do when it is not, from simple positioning to clearing a choking adult. Entry 2.3 explains how to move air for a patient who cannot, with a bag-valve-mask and supplemental oxygen. Entry 2.4 turns to the lungs and the

conditions that make breathing hard. Entry 2.5 ends with anaphylaxis, which can close an airway within minutes and which a bystander may be the first to treat.

Everything here assumes the habits of Part 1: a safe scene, a first look, and a recheck after every action. As before, what you may do depends on your scope of practice and local protocol. Where a skill needs training that a book cannot give, the text says so plainly and tells you when to stop and call for help.

2.1 Airway Anatomy and Assessment

The airway is the passage that carries air from the outside to the lungs. You cannot judge whether it is open, or why it has closed, without a picture of what it is made of. This entry gives you the picture and then the short set of observations that tell you how well it is working.

WHAT IT IS. Air enters through the nose and mouth and passes into the throat, called the pharynx. Below the throat sits the larynx, or voice box, guarded by a flap of tissue called the epiglottis that covers the entrance when you swallow. Below the larynx the windpipe, or trachea, runs down into the chest and divides into the two main bronchi, which branch into ever smaller tubes inside the lungs. The part above the larynx is the upper airway, and most emergency airway care is aimed there, because it is the part you can reach and open.

Figure 5 shows these structures and the landmarks you use.

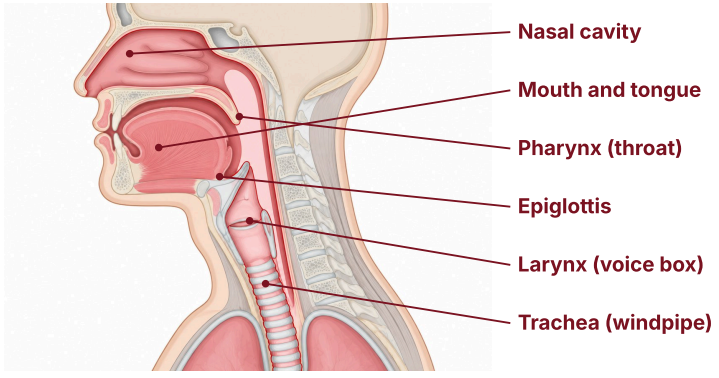


Figure 5. The upper airway from nose and mouth to the windpipe, with the structures that most often narrow or block it.

WHY IT MATTERS. The upper airway is narrow, shared with the route for food and drink, and held open by muscle. Anything that weakens the muscle, adds swelling, or puts something in the way can close it. Because the brain depends on a constant supply of oxygen, a blocked airway can cause loss of consciousness and then cardiac arrest if it is not cleared.

WHAT NARROWS OR BLOCKS IT. Five causes account for most emergencies. In an unconscious patient lying on the back, the tongue and soft tissues of the throat can fall backward and close the passage, and this is the commonest cause of obstruction in a patient with a depressed level of consciousness. Food or a foreign object can lodge in the throat. Fluid, vomit or blood can fill the airway. Swelling from an allergic reaction, infection, a burn or an injury can narrow it from within. A direct injury to the face or neck can distort it. Each cause points to a different first action, which is why the next entry is organized by cause and not by technique.

ASSESSING THE AIRWAY. Look, listen and feel, as in the primary survey. Ask the patient a simple question. A patient who answers in a normal voice has an open airway for the moment. Then

watch the chest and abdomen, listen for sounds, and feel for air at the mouth and nose.

What airway findings suggest

Finding	What it suggests
Normal voice, full sentences	Airway open for now
Hoarse or whispered voice	Swelling or narrowing of the throat
High-pitched sound on breathing in (stridor)	Significant partial blockage
Gurgling	Fluid in the airway
Chest and abdomen moving in opposite directions	Obstruction with strong effort to breathe
No air felt or heard at the mouth or nose	Complete obstruction or no breathing

Airway obstruction is an emergency. In a partial obstruction, air still moves but less than it should and often with noise. In a complete obstruction, no air moves, the patient cannot speak or cough, and the chest and abdomen may move in opposite directions as the patient strains. A blue or gray tint to the lips, called central cyanosis, is a late sign. Do not wait for it.

RED FLAGS

A very hoarse or whispered voice, a high-pitched sound on breathing in, drooling or inability to swallow, swelling of the face or tongue, a patient who cannot speak or cough, or a blue or gray tint to the lips. Call for help at once.

NURSING AND RESPONDER PRIORITIES. Assess the airway first and again after every change in the patient, treat obstruction as an emergency, call for help early, and do not delay action while waiting for a blue tint to appear.

See also: Entry 1.3 (primary survey); Entry 2.2 (opening and clearing the airway); Entry 6.1 (child airway differences).

2.2 Airway Obstruction and Basic Airway Management

The simple measures in this entry can be done with few or no tools, and they solve most airway problems you will meet before advanced help arrives. They follow the cause. If the tongue has fallen back, you reposition. If there is fluid, you suction. If there is a solid object, you relieve choking. If there is swelling, you call for help and treat the cause, as Entry 2.5 describes for allergic swelling.

WHAT IT IS. Basic airway management is the set of non-invasive actions that open and protect the airway without placing any device inside the patient. It includes positioning the head and jaw, clearing the mouth with suction, placing a patient who is breathing but unresponsive on their side, and relieving choking.

WHY IT MATTERS. Each measure is quick and each can be done by a student with the right training and a supervisor present. They matter because they act on the cause directly and because they must be repeated: an airway opened once may close again as the patient's condition changes.

OPENING THE AIRWAY. The two standard positions lift the tongue away from the back of the throat. In the head-tilt chin-lift, you tilt the forehead back with one hand and lift the chin with the fingertips of the other. It is used for a patient with no suspected

neck injury. In the jaw thrust, you place your fingers behind the angles of the jaw on both sides and lift the jaw forward without tilting the head. It is the choice when a neck injury is possible, because it moves the neck less. If the jaw thrust does not open the airway, your protocol may direct you to use the head tilt anyway, since an open airway comes first. Entry 4.3 returns to the neck.

Figure 6 compares the two positions.

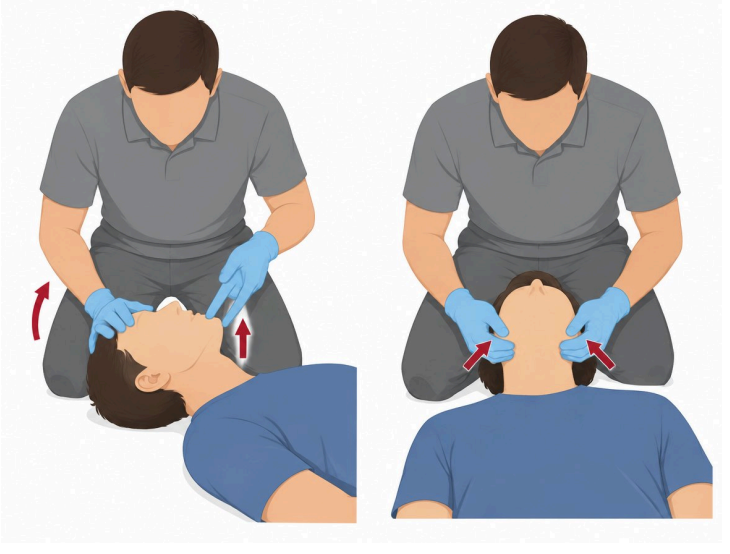


Figure 6. The head-tilt chin-lift and the jaw thrust, drawn side by side to show where the hands go.

CLEARING THE MOUTH. Suction removes liquid and small particles from the mouth and throat. Use it only when you can see or hear fluid, apply it in short bursts and not continuously, and stop to check the patient's breathing and condition between bursts, because suction also removes air. Follow the device maker's instructions and your protocol for how long and how deep.

PROCEDURE: OPEN THE AIRWAY OF AN UNRESPONSIVE PATIENT

Purpose: to keep air moving past the tongue. Scope: follows your training level and local protocol.

1. Make the scene safe, put on gloves, and call for help.
2. Check for a response and for normal breathing (Entry 1.3).
3. If you suspect a neck injury, use the jaw thrust; otherwise, tilt the head and lift the chin.
4. Look in the mouth and clear anything you can see and reach, such as fluid or loose debris, with suction if it is available and you are trained.
5. Look, listen and feel for breathing again after the airway is open.
6. If the patient is breathing normally and not injured, and you must leave or cannot stay with them, place them on their side in the recovery position; otherwise keep them on the back with the airway held open.
7. If the patient is not breathing normally, begin CPR (Entry 3.4).
8. Reassess and keep checking breathing until help takes over.

> **⚠ SAFETY**
>
> Do not put your fingers blindly into a patient's mouth. Never leave an unresponsive patient alone, and do not move a patient with a possible neck injury unless the scene is unsafe or you must move them to give life-saving care, as your protocol directs.

STOP AND CALL FOR HELP IF the airway does not open, if suction does not clear the mouth, or if the patient stops breathing.

These steps work in a fixed order because each one answers what the last one found. Positioning solves the common problem first, the fallen tongue. Clearing the mouth solves the second most common. Checking breathing then tells you whether the patient needs only an open airway or needs breathing support as well.

THE RECOVERY POSITION. A patient who is breathing normally but is unresponsive, and who has no suspected neck or spinal injury, is at risk from their own tongue and from vomit. Placing them on their side lets fluid drain from the mouth and keeps the tongue forward. Kneel beside the patient, place the arm nearest you at a right angle to the body, bring the far arm across the chest and hold the back of that hand against the near cheek, bend the far knee, and roll the patient toward you by pulling on that knee. Adjust the top leg so that the hip and knee are bent at right angles and tilt the head back slightly so the airway stays open. Check breathing again at once and keep checking. The position is for a patient who is breathing normally; a patient who is not breathing normally needs CPR, not repositioning, and a patient with a possible spinal injury is moved only as your protocol directs.

CHOKING IN AN ADULT. Choking is a solid object blocking the airway. A patient with a partial blockage can still cough, speak or breathe with effort, and you should encourage coughing and stay close without hitting them. A patient with a severe blockage cannot speak, cough or breathe, may clutch the throat, and may turn blue or gray. That patient needs immediate help. Ask “Are you choking?” and, if they nod and cannot speak, act.

PROCEDURE: RELIEVE CHOKING IN A CONSCIOUS ADULT

Purpose: to dislodge an object that blocks the airway. Scope: follows your training level and local protocol; the exact sequence your service teaches takes precedence.

1. Ask "Are you choking?" and check whether the person can cough, speak or breathe.
2. Send someone to call emergency services, or call for help yourself as soon as you can.
3. Stand to one side and slightly behind the person, support the chest with one hand, and lean them forward.
4. Give up to five firm blows between the shoulder blades with the heel of your other hand.
5. If the object is still stuck, stand behind the person, place a fist just above the navel, and give up to five quick inward and upward abdominal thrusts.
6. Keep alternating back blows and abdominal thrusts until the object comes out or the person stops responding.
7. If the person becomes unresponsive, lower them carefully to the ground, call for help, and begin CPR (Entry 3.4).
8. Even if the object comes out, the person should be checked by a clinician.

> **⚠ SAFETY**
>
> Abdominal thrusts can injure internal organs. For a person who is visibly pregnant or too large to reach around, your training may direct chest thrusts instead. Do not use abdominal thrusts on infants; Entry 6.2 covers them.

NURSING AND RESPONDER PRIORITIES. Match the action to the cause, open the airway first, suction only when you can see or hear fluid, relieve choking in the order you were taught, begin CPR

if a choking patient becomes unresponsive, never leave an unresponsive patient alone, and recheck the airway after every move.

See also: Entry 1.3 (primary survey); Entry 3.4 (CPR); Entry 4.3 (spinal motion restriction); Entry 6.2 (infant and child choking).

2.3 Ventilation Support: Bag-Valve-Mask and Oxygen Delivery

Sometimes the airway is open and the patient still is not moving enough air. Breathing may be too slow, too shallow or absent. At that point a trained responder moves air for the patient with a bag-valve-mask, often connected to oxygen. This entry explains the device, the steps, and the oxygen devices you may meet. It gives device types and when they are used, and no flow-rate figures.

WHAT IT IS. A bag-valve-mask, written BVM, has three parts: a soft face mask that seals over the nose and mouth, a one-way valve, and a self-refilling bag that you squeeze to push air into the lungs. Released, the bag refills from room air or from an attached oxygen supply. It is used on patients who are not breathing or are breathing too weakly, and it works best with two trained people, one to hold the mask and one to squeeze the bag.

WHY IT MATTERS. Moving air by hand is a skill that looks easy and is not. A poor seal lets air escape. Squeezing too hard or too fast can push air into the stomach and cause vomiting. A technique that moves the chest well on a manikin may fail on a patient with a beard, a small mouth or a swollen face. Practice on manikins, under supervision, is what turns a clumsy first attempt into a reliable one.

USING A BAG-VALVE-MASK. Choose a mask that covers the nose and mouth without pressing on the eyes. With two people, one holds the mask with both hands, keeping the airway open

with the jaw thrust or head tilt, while the other squeezes the bag. With one person, the single hand and the bag are harder to manage, and the seal is the usual point of failure. Squeeze gently and just enough to make the chest rise, then release and let the chest fall. Watching the chest, not counting squeezes, tells you whether air is going in.

Figure 7 shows the two-hand mask seal.



Figure 7. A two-hand mask seal with a second person squeezing the bag.

For an adult who has a pulse but is not breathing normally, a common rule is one breath about every six seconds, which is about ten breaths a minute. Your protocol sets the rate you follow, and the rate for a child or infant differs.

PROCEDURE: ASSIST BREATHING WITH A BAG-VALVE-MASK

Purpose: to move air into the lungs of a patient who is not breathing adequately. Scope: follows your training level and local protocol; usually done under supervision.

1. Make the scene safe, put on gloves and eye protection, and call for help.
2. Open the airway and clear the mouth (Entry 2.2).
3. Check for breathing and a pulse; if there is no pulse, begin CPR (Entry 3.4).
4. Choose a mask that covers the nose and mouth, and attach the bag and, if available, the oxygen supply.
5. Hold the mask on the face with both hands, with the jaw lifted, so that no air leaks.
6. Have the second person squeeze the bag gently until the chest begins to rise, then release.
7. Give breaths at the rate your protocol sets, watching the chest rise and fall.
8. Keep the airway open, reassess often, and stay with the patient until help takes over.

> ⚠ **SAFETY**
>
> Do not over-squeeze or give breaths too fast. If the stomach swells or the patient vomits, turn the patient on the side if you can, clear the mouth, and call for help. Never use oxygen near flame, sparks or smoking.

STOP AND CALL FOR HELP IF the chest does not rise, the patient vomits, or you cannot make a seal.

IS THE VENTILATION WORKING? Judge each breath by its effect, not by the count. The chest should rise gently and fall fully as the bag is released. You should feel little resistance in the bag and see no air escaping around the mask. A good response may include improved skin color and, in a patient with a pulse oximeter, a rising reading, though that takes time. If the chest does not rise, reposition the head and jaw, reseat the mask, and check for an obstruction before squeezing harder. Increasing force is almost never the answer.

OXYGEN DELIVERY DEVICES. Oxygen can be given by several devices, chosen by how much the patient needs and by the patient's ability to breathe. A nasal cannula is a soft tube with two short prongs that rest in the nostrils, used for patients who are breathing well and need a small boost. A simple face mask covers the nose and mouth and is used when a moderate increase is needed. A non-rebreather mask has a bag, called a reservoir, that fills with oxygen so that the patient breathes in almost pure oxygen, and it is used for the sickest breathing patients. A BVM can be connected to oxygen and to a reservoir to deliver the highest concentration to a patient who cannot breathe for themselves. Your protocol and medical direction set the choice, the flow, and the target for each patient. Giving oxygen is part of the care of many roles, but giving medicines is outside many roles; follow your scope and medical direction.

NURSING AND RESPONDER PRIORITIES. Open and clear the airway before ventilating, make a firm seal, give gentle breaths until the chest rises, use two people when you can, keep oxygen away from flame, and recheck after every change.

See also: Entry 2.2 (airway); Entry 3.4 (CPR); Entry 6.1 (child sizing and rate).

2.4 Respiratory Distress: Asthma, COPD and Pneumonia

Breathing difficulty is one of the commonest reasons people call for help, and its causes range from a minor chest cold to a life-threatening attack. This entry gives you a way to judge how serious it is and covers three common causes.

WHAT IT IS. Respiratory distress is difficult or labored breathing. Respiratory failure is the stage at which breathing can no longer meet the body's needs, and it can follow distress quickly, particularly in a tiring patient. Signs of distress include sweating, using the neck and shoulder muscles to breathe, belly breathing, and a gray or blue tint to the lips. A rate that is high or rising is a warning. Signs of failure include drowsiness or confusion, a rate that is slowing in a tiring patient, a very quiet chest in a patient who was wheezing, and blue lips.

WHY IT MATTERS. A patient who is working hard to breathe can tire, and a tiring patient can stop. The signs of tiring are subtle: the patient becomes calmer and quieter, which is easy to mistake for improvement. Reassessing breathing rate, effort and mental state at short intervals is what catches the change.

REASSESSING A BREATHING PATIENT. Because the patient's condition can change quickly, set a rhythm for rechecking. Each time, note the breathing rate, how many words the patient can say at once, whether the neck and shoulder muscles are working, the color of the lips and the level of alertness. Write each set down with the time, as Entry 1.5 describes. A patient who needs more help than before is worsening even if each single number looks acceptable. A patient who is tiring will often change position, lean forward on outstretched arms and start to say fewer words. Let the pattern guide your call for help.

ASTHMA. Asthma is a long-term condition in which the airways narrow, tighten and become inflamed in response to triggers such as allergens, infection, exercise or cold air. In an attack, the patient may wheeze, cough, feel tightness in the chest, and struggle to breathe. Many patients carry a reliever inhaler containing albuterol (salbutamol). Giving medicines is outside many roles; follow your scope and medical direction. A severe attack is shown by speaking only a few words at a time, a high breathing rate, visible effort, a very quiet chest, drowsiness, or no improvement after the patient's usual reliever. Sit the patient upright, keep them calm, call for help, and help them use their own inhaler as they have been taught and as your protocol allows.

COPD. Chronic obstructive pulmonary disease, written COPD, is a long-term lung disease, often caused by smoking, in which the airways are narrowed and the lungs are damaged. Patients are short of breath on effort, cough, and bring up mucus, and they have flare-ups called exacerbations. Oxygen needs special care here. In some patients with COPD, a high concentration of oxygen can depress breathing, so these patients need a lower oxygen target set by protocol or medical direction. Give oxygen as directed and watch breathing closely; do not withhold oxygen from a patient who needs it.

PNEUMONIA. Pneumonia is an infection of the lungs. It causes fever, cough, sometimes chest pain on breathing in, and a fast, shallow breathing rate. In older adults or very ill patients it may cause confusion without much cough or fever. Pneumonia can lead to sepsis, which Entry 5.4 covers. First priorities are an upright or comfortable position, oxygen as directed, early reassessment, and a call for help if breathing is hard, the patient is confused, or the lips are blue.

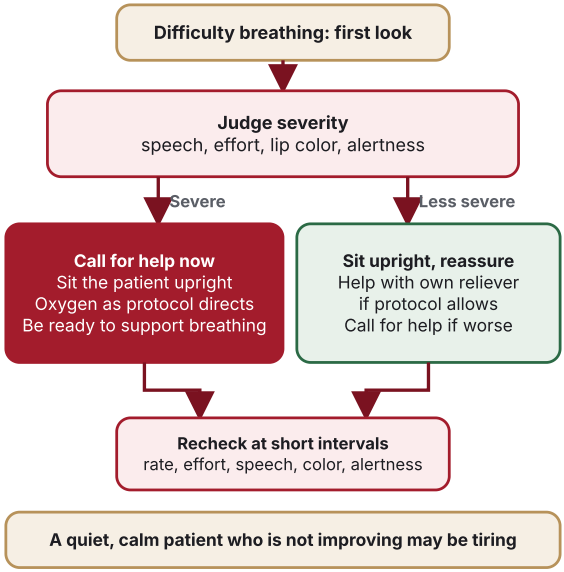


Figure 8. A first-priority flow for a patient with difficulty breathing: judge the severity, call for help, position the patient, and recheck.

RED FLAGS

Speaking only a few words at a time, a very fast or slowing breathing rate, blue or gray lips, drowsiness or new confusion, a chest that has gone quiet, or a patient who becomes calmer without improving. Call for help at once.

SCOPE NOTE

Helping a patient use their own prescribed inhaler, and giving oxygen, depend on your role, your protocol and medical direction. If you are not sure whether you may, ask a supervisor, and always call for help in a severe attack.

NURSING AND RESPONDER PRIORITIES. Judge severity by effort, speech, color and mental state, position the patient upright, call early for help, give oxygen as your protocol directs with care for patients who have long-term lung disease, and reassess often because a quiet patient may be a tiring one.

See also: Entry 1.3 (primary survey); Entry 2.3 (oxygen devices); Entry 5.1 (confusion with poor breathing); Entry 5.4 (sepsis); Entry 6.1 (children).

2.5 Anaphylaxis

Anaphylaxis is a severe allergic reaction that affects the whole body and can be fatal. It tends to arrive fast, often within seconds or minutes of contact with the trigger, and it can close the airway, drop the blood pressure, or both. A bystander with no medical training may be the first person to treat it, which is why this short entry matters more than its length suggests.

WHAT IT IS. In anaphylaxis, the immune system reacts to a substance it has become sensitive to, such as an insect sting, a food or a medicine, and releases chemicals that cause the airways to tighten and the blood vessels to widen. Common triggers are foods, insect stings and medicines. Some people have a reaction with no known cause.

WHY IT CAN KILL. Two things fail in anaphylaxis, often together. Swelling and tightening narrow the airway, so air cannot get in. Widened blood vessels leak fluid and drop the blood pressure, so the heart cannot keep the brain supplied. Either can cause collapse. Because the reaction can recur after it seems to have settled, a patient who improves still needs to be assessed by a clinician.

RECOGNITION. Symptoms can involve several body systems at once. They include hives, itching or redness of the skin, swelling of the face, eyes or tongue, difficulty breathing, wheezing, a tight chest, a hoarse or whispered voice, difficulty swallowing, abdominal pain, vomiting, dizziness or fainting, and a sense of doom. A warning sign of dangerous throat swelling is a very hoarse voice or coarse noise on breathing in. When a person has a known allergy and develops breathing difficulty, throat swelling or faintness soon after exposure, treat it as anaphylaxis.

FIRST PRIORITIES. Anaphylaxis is an emergency. Call emergency services immediately. Check airway, breathing and circulation, and be ready to start CPR. Remove the trigger if you can, for example by scraping off a bee stinger. If the person has a prescribed epinephrine (adrenaline) auto-injector, help them use it at once; the first-line treatment is epinephrine, and delay is dangerous. Giving medicines is outside many roles; follow your scope and medical direction.

PROCEDURE: HELP A PATIENT USE THEIR OWN EPINEPHRINE AUTO-INJECTOR

Purpose: to help the patient receive their own prescribed, premeasured epinephrine as the device instructions direct. Scope: follows your training level, state law and local protocol.

1. Call emergency services, or ask someone to call, and stay with the patient.
2. Ask whether they carry an auto-injector, and find it.
3. Read the device label and follow its printed instructions exactly; do not guess or improvise.
4. Help the patient or, where your role and the law allow, use the device on the patient as the label shows.
5. Note the time the injection was given and tell the responders.

6. Place the patient in the position that suits their breathing: lying flat with legs raised if they are faint, or sitting up if breathing is hard.
7. Keep checking breathing and consciousness, and be ready to start CPR.
8. Even if the patient improves, they need to be assessed by a clinician; symptoms can return.

> **⚠ SAFETY**
>
> Do not give anything by mouth if the person has trouble breathing. Do not put a pillow under the head of a person who has trouble breathing. Do not assume earlier allergy shots give full protection. Used needles must go in a sharps container; keep your fingers away from the needle end of the device.

STOP AND CALL FOR HELP IF breathing becomes difficult, the voice changes, swelling spreads, or the patient becomes drowsy or faint.

The steps follow the logic of the problem. Anaphylaxis is a race between a reaction that is accelerating and a treatment that takes effect in minutes, so the first moves are to bring help and the treatment together. Position protects the airway and the blood pressure, and the notation of time lets the next team judge what to do.

NURSING AND RESPONDER PRIORITIES. Recognize the pattern early, call for help at once, help the patient use their own epinephrine auto-injector as the device directs, position the patient for breathing or faintness, never give food or drink, and keep reassessing until a clinician takes over.

See also: Entry 1.3 (primary survey); Entry 2.2 (airway); Entry 3.5 (shock); Entry 6.1 (children).

Illustrative Case

ILLUSTRATIVE CASE: NOT A REAL PATIENT

Marcus was a first-year paramedic student helping at a weekend youth camp with the camp nurse, Dana. During an afternoon game, a fourteen-year-old camper named Leo sat down on the grass and said he could not catch his breath. Marcus reached him first, checked that the area was clear, and put on gloves.

Leo was sitting forward, resting his hands on his knees. He could say only a few words between breaths and was breathing fast. Marcus could hear a high wheeze. Leo's lips looked normal, and he was alert. Leo said he had asthma and that his inhaler was in his bag at the cabin, and that he had been running hard in the cold wind.

Marcus told Dana what he saw while she fetched the inhaler and the camp's first-aid kit. He kept Leo sitting upright, spoke slowly, and counted his breathing rate again two minutes later. It had not changed. Dana arrived, helped Leo use his own inhaler as he had been taught, and told Marcus to keep watching his color and how much he could say. When Leo said a full sentence, they relaxed a little. Dana still called for the camp doctor, because the breathing had not settled and a night in the cabin would be unsafe without an assessment.

Care Priorities

CARE PRIORITIES

When breathing is failing, the airway comes before anything else, and a patient who cannot get air in cannot be helped by a device that moves air out. Position before equipment: a tongue that has fallen back needs a hand, not a machine. A patient who is working hard to breathe needs help before they tire, so treat effort as a warning even when the oxygen reading looks fair. When a trigger is known, remove it, but never at the cost of delaying a call for help. In anaphylaxis, bring the call and the treatment together, because every minute of delay lets the reaction advance. If a skill in this Part is outside your scope, say so and keep the patient's airway open until someone trained arrives.

Review Questions

1. A patient's chest and abdomen move in opposite directions and no air can be felt at the mouth. What does this suggest?
   - **A)** Normal breathing.
   - **B)** Complete airway obstruction.
   - **C)** A mild chest cold.
   - **D)** Anxiety only.

2. Which position is preferred for opening the airway of an unresponsive patient with a possible neck injury?
   - **A)** Head-tilt chin-lift.
   - **B)** Sitting upright.
   - **C)** Jaw thrust.
   - **D)** Face down.

3. When should you use suction?
   - **A)** At all times during airway care.
   - **B)** When you can see or hear fluid in the mouth or throat.
   - **C)** Only after the patient wakes up.
   - **D)** Before checking for a response.

4. A conscious adult cannot speak, cough or breathe and points to the throat. Which action is correct?
   - **A)** Give a drink of water.
   - **B)** Wait for the person to improve.
   - **C)** Follow your training for back blows and abdominal thrusts.
   - **D)** Lay the person flat and begin chest compressions at once.

5. Which is a risk of squeezing a bag-valve-mask too hard or too fast?
   - **A)** The mask will fall off.
   - **B)** The oxygen will run out.
   - **C)** Air can enter the stomach and cause vomiting.
   - **D)** The patient will wake up.

6. A patient with severe asthma becomes quieter and calmer but still has blue lips and cannot speak more than a word. What does this suggest?
   - **A)** The attack is improving.
   - **B)** The patient is tiring and may be failing.
   - **C)** The patient is faking.
   - **D)** Nothing needs to change.

7. After a bee sting, a person develops hives, a hoarse voice and difficulty breathing. What is the first priority?
   - **A)** Take a photograph of the sting.
   - **B)** Give food to restore strength.
   - **C)** Call emergency services and help with the patient's own epinephrine auto-injector.
   - **D)** Wait to see whether the symptoms pass.

8 Short case. A patient with long-term lung disease has become short of breath at home and is drowsy. List, in order, the first four things you would do and explain why oxygen needs care in this patient.